A Robust and Runtime-Efficient Track-to-Track Fusion for Automotive Perception Systems

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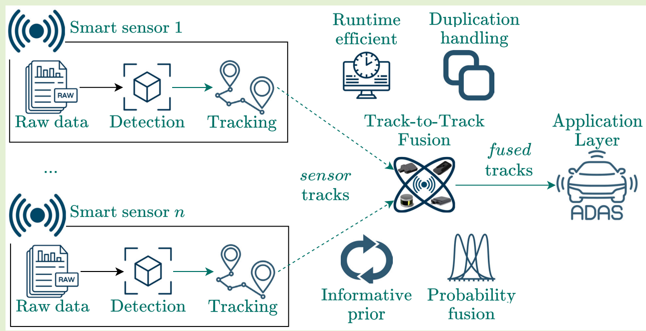
***Abstract*—Advanced driver assistance and autonomous systems require an enhanced perception system, fusing the data of multiple sensors. Many automotive sensors provide high-level data, such as tracked objects, i.e., *tracks*, usually fused in a track-to-track manner. The core of this fusion is the track-to-track association (T2TA), intending to create assignments between the tracks. In conventional T2TA, the assignment likelihood function is derived from "diffuse prior," neglecting that the sensors may provide duplicated tracks of a target. Moreover, conventional association algorithms are usually computationally demanding due to the combinatorial nature of the problem. The first motivation of this work is to obtain a computationally efficient and robust solution, reflecting these problems. Another crucial element of track-to-track fusion is track management, which maintains the list of tracks by initializing and deleting tracks, thus having a great impact on the reliability of the fusion output. In this article, we propose a novel track-to-track fusion architecture in which the fused tracks are fed back to the association. The proposed method comprises two main contributions. First, a computationally efficient association algorithm is provided in which the "diffuse prior" is replaced with an informative prior, exploiting the feedback loop of the fused tracks. Moreover, it tackles duplicated tracks. Second, a track management system (TMS) relying on a revamped track existence probability fusion is proposed, contributing to efficient false track filtering and continuous object tracking. The proposed methodology is evaluated on real-world data of a frontal perception system. The results show that the proposed association outperforms the conventional methods; still, it maintains a favorable complexity, contributing to real-time applicability. The TMS relying on the revamped existence probability fusion can efficiently filter false tracks and continuously track objects. Moreover, the resulting overall track-to-track fusion outperforms the state-of-the-art multiobject tracking-based fusion algorithms.**



***Index Terms*—Advanced driver assistance systems (ADAS), environment perception, radar tracking, sensor fusion, smart cameras, track-to-track fusion.**

I. Introduction

THE increasing level of vehicle automation promises many positive changes in sustainability, economics, and traffic engineering [1], [2]. Advanced driver assistance systems (ADAS) have already had a great impact on road safety [3]. However, the reliability of these systems has a significant impact on user acceptance [4]. An important aspect of reliability is managing false alarms, which often stem from false detections of the perception module [5], [6]. Sensor data fusion has a crucial role in the reliability of the perception, as it aims to compensate for the disadvantages of the different types of sensors [7], [8]. Fusion can be performed on different abstraction levels of data in various architectures [9], [10], [11], which highly impacts its performance. However, many automotive sensors provide high-level data, such as object detections or tracked objects, *tracks*. Object detections are usually fused in a central tracking algorithm, which updates a common track list with the detections of the different sensors relying on a Kalman filter, as in [12] and [13]. Local tracks of the sensors can also be fused in a central tracking architecture, considering them as measurements in a second Kalman filter. However, it raises the correlation problem since the sensor tracks generated by the same target are not independent due to the common process noise [14]. Therefore, track-to-track methods, which aim to consider the correlation problem, are often used to combine tracks provided by automotive smart sensors [15]. Besides the vehicle-level perception, track-to-track fusion can

Received 12 February 2025; accepted 4 March 2025. Date of publication 13 March 2025; date of current version 16 April 2025. This work was supported in part by European Union within the framework of the National Laboratory for Autonomous Systems under Grant BME-NVA-02; in part by the Ministry of Innovation and Technology of Hungary from the National Research, Development and Innovation Fund, through the TKP2021 Funding Scheme under Grant RRF-2.3.1-21-2022-00002; and in part by the Ministry of Culture and Innovation of Hungary from the National Research, Development and Innovation Fund, through the EKÖP-24-3 Funding Scheme under Grant 2024-2.1.1-EKÖP. The associate editor coordinating the review of this article and approving it for publication was Prof. Huang Chen Lee. *(Corresponding author: Tamás Bécsi.)*

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Digital Object Identifier 10.1109/JSEN.2025.3548899

contribute to cooperative perception systems [16] and other distributed sensor networks in various applications [17], [18].

A. Related Works

Track-to-track fusion can be implemented following two main architectures, the sensor-to-sensor and sensor-to-global, usually comprising three modules: association, state and covariance fusion, and track management [19]. The two architectures differ in the core module of the fusion, the association, answering the question of which tracks belong to the same target. In sensor-to-sensor architecture, the sensor tracks are assigned to each other, clustering them so that those generated by the same target are included in the same cluster, as in [20], In sensor-to-global architecture, the sensor tracks are assigned to the fused (global) tracks of the previous cycle, feeding them back to the association [21].

The most challenging task of track-to-track fusion is the association conventionally tackled by statistical reasoning, judging whether two tracks are generated by the same target relying on probability calculation and hypothesis testing [22]. In a sensor-to-sensor configuration, the assignment likelihood of two tracks, quantifying how likely they were generated by the same target, was derived in [23]. This likelihood function was generalized for multiple sensors in [24] and [25], neglecting and considering dependent errors, respectively. While, in the presence of systematic sensor errors (bias), the likelihood was given in [26]. Having the association likelihood of the set of track pairs, the assignment problem is solved by maximizing the likelihood. In the case of two sensors, this simplifies to an auction problem, which can be solved by the Hungarian method [27]. However, its $\mathcal{O}(n^3)$ complexity already makes the real-time applications challenging, particularly in dense urban environments. In the case of multiple sensors, this forms an integer linear programming problem that is generally NP-hard. Moreover, these methods are derived, assuming a "diffuse," noninformative prior. In other words, the association does not have prior information about the tracked targets. Prior association hypotheses are exploited in [28] similar to multiple hypotheses tracker (MHT) [29] to preserve the continuity of the fused tracks. However, just like MHT, this can be memory and computationally demanding. In sensor-to-global architecture, thanks to the feedback, the likelihoods of the sensor tracks are computed given to the global tracks instead of diffuse prior. Then, the assignment problem can be solved by the Hungarian method sequentially for each sensor [30]. However, having numerous sensors may compromise real-time running due to the $\mathcal{O}(n^3)$ complexity mentioned above.

Thus, research toward runtime-efficient heuristic methods already exist. For example, in statistical reasoning methods, iterative greedy solutions are used to solve the assignment problem after computing the assignment likelihoods in both sensor-to-sensor [31] and sensor-to-global architectures [21], replacing the computationally heavy optimization. Besides their complexity, the statistical reasoning methods have several performance limitations. First, they require a user-defined threshold, the cost of the "unassignment," which may significantly impact the performance of the association. Moreover, they assume idealistic sensors that provide at most one track per target. While real sensors, particularly radars, often duplicate targets. Another approach to tackle track-to-track association (T2TA) is fuzzy modeling [32], [33], [34]. Although these methods have low-computational costs, they still rely on user-defined thresholds. In [35], an adaptive threshold is used to decrease the sensitivity to initial thresholds. In recent years, deep learning-based track association methods have also come to the fore with the aim of reducing the computational cost and eliminating the dependency on prior assumptions and thresholds. The association problem is transformed into a binary classification problem in [36] solved by AdaBoost and a decision tree, while a convolutional neural network is used in [37]. A track fusion and track segmentation (TF-TS) method is proposed in [38], generating the association matrix in an end-to-end manner relying on a long short-term memory (LSTM) network. However, deep learning-based methods have not become widespread in automotive applications yet due to safety concerns.

After determining which tracks are generated by the same target, they are combined by fusing their state and covariance estimates. The first fusion formula was derived subsequently from [14] in [39] for the case of two sensors and generalized to the case of n-sensors in [40]. However, in [41], it is shown that [39] is only optimal in maximum likelihood (ML) sense and not in terms of mean-squared error (mse), proposing another formula, called the information matrix fusion (IMF), adapted to a sensor-to-global configuration in [42]. Besides IMF, there are three algorithms often used in automotive applications in sensor-to-sensor configuration: the covariance intersection, covariance union, and the use of cross-covariance. A comparison of these methods, including an asynchronous Kalman filter, is provided in [43], showing that the use of cross-covariance outperforms the others.

The last step of track-to-track fusion is track management, which intends to maintain the list of global tracks by handling appearing and disappearing objects, i.e., initiating new tracks and deleting outdated ones [44]. As sensors may provide false tracks, new tracks are usually initiated *tentatively*, keeping them only internally. A track is only transmitted to the application level when it is *confirmed* after some cycles [45]. Besides track initiation and deletion, track management has to deal with track confirmation; therefore, it has a significant role in false track filtering. Despite its impact on the reliability of the perception system, track management is a less-covered topic of track-to-track fusion. In automotive multiobject tracking, three strategies can be found in the literature: the M/N logic, sequential probability ratio test (SPRT), and existence probability-based [46]. As the simplest method is arguably the M/N logic, it is also used in track-to-track fusion [30]. However, since the built-in tracking algorithm of the sensors, such as integrated probabilistic data association (IPDA) [47] or joint IPDA (JIPDA) [48], usually integrates an existence probability estimation method, there are obvious reasons behind fusing these estimates of the local sensor tracks. In [49] and [50], IPDA and JIPDA are specifically adapted to fuse

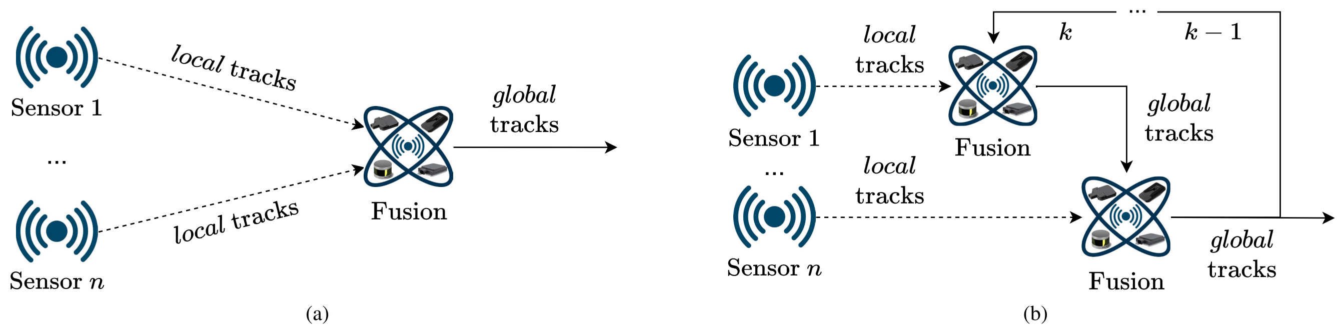

Fig. 1. Track-to-track fusion architectures. (a) Sensor-to-sensor architecture. (b) Sensor-to-global architecture.

the existence probabilities of the sensor tracks in a Bayesian manner. However, since it does not characterize uncertainties well, a novel method relying on Dempster–Shafer (DS) belief fusion [51] is proposed to fuse the existence probability of the tracks in sensor-to-sensor [52] and sensor-to-global [21] configurations.

B. Contributions of the Article

In this article, a hybrid track-to-track fusion architecture is proposed comprising a novel T2TA and track management system (TMS). The novelties and benefits of the proposed T2TA over prior works include the following.

1) The "diffuse prior" of the conventional sensor-to-sensor T2TA likelihood function is replaced with informative prior, exploiting the feedback of the hybrid architecture.
2) It deals with duplicated tracks.
3) The cost of "unassignment" is computed from standard and estimable sensor parameters.
4) It maintains a favorable runtime complexity.

While the following points outline the leverages of the TMS.

1) It combines the benefits of sensor-to-sensor and sensor-to-global architectures.
2) It implements a novel operator that fuses the existence probability of the tracks, reflecting the criticism of DS theory (DST) in such cases as fusing belief estimates from multiple sources integrated in a cumulative manner [53].

Thanks to the novel T2TA and TMS, the proposed methodology significantly outperforms state-of-the-art track-to-track fusion algorithms, providing a balanced tracking and false alarm filtering performance.

The rest of the article is organized as follows. Section II gives a deeper overview of the theoretical background of track-to-track fusion. Section III details the proposed methodology. The experimental setup and the results are demonstrated in Sections IV and V. Finally, the conclusions are drawn with an outlook on future works in Section VI.

II. Theoretical Background

Track-to-track fusion can be implemented in different architectures [19]. The two most common approaches, the sensor-to-sensor and sensor-to-global, are illustrated in Fig. 1. In a sensor-to-sensor architecture, the *local* tracks of the sensors are fused with each other, following a feed-forward structure, as shown in Fig. 1(a). While the local sensor tracks are fused with the *global* tracks in a sensor-to-global architecture. This means that the fusion has a memory, feeding the resulting global tracks back to the next cycle, as seen in Fig. 1(b). In some cases, the global tracks are fed back to the sensors to improve the performance of their local, built-in tracking algorithms [50]. However, in automotive applications, this architecture is less frequent due to the challenges and cost implications of two-way communication.

The feed-forward architecture makes sensor-sensor operation the biggest advantage of being easy to implement. However, besides sensor-to-sensor fusion requires synchronized sensor data, sensor-to-global architecture has several advantages over sensor-to-sensor. First, thanks to memory, the global tracks can be distinguished and referred to by their unique identifiers, which can help select relevant targets for such ADAS applications as automated emergency braking (AEB) and adaptive cruise control (ACC). Moreover, maintaining the state of a global track with memory can provide a smooth transition between the field of views (FoVs) of the sensors. However, sensor-to-global architecture has to deal with some challenges. Although it can work asynchronously, updating the common global track list whenever a sensor transmits its data, out-of-sequence data must be handled [54]. The correlation problem must also be considered, as the state estimates of a local track are not independent over time [42]. Moreover, the local sensor and global system track of the same target also have dependent errors due to the common process noise [14]. Still, sensor-to-global architecture is more widespread in automotive applications. The architecture also has significant implications on the association and track management algorithms that are detailed in Sections II-A and II-B. It must be noted that state and covariance fusion also has a crucial role. Still, it is out of the scope of this theoretical introduction as the proposed method contributes to association and track management. The theoretical background of state and covariance fusion is provided in [43].

A. Track-to-Track Association

The T2TA is intended to determine which tracks are generated by the same target, creating assignments between the local sensor tracks or local and global system tracks, depending on the architecture. As seen in Fig. 1(b), in sensor-to-sensor architecture, the local tracks are assigned to each other, forming clusters from the tracks assumed to be generated by the same target. Therefore, idealistically, clusters may comprise as many tracks as many sensors can detect the corresponding target. Supposing two sensors, i and j, tracking an x_t target, let $\hat{x}_{t_i}$, $\hat{x}_{t_j}$, P_{t_i}, and P_{t_j} denote the state estimates and covariances of the local $\mathbf{x}_{t_i}$ $(t_i = 1, \ldots, N_i)$ and $\mathbf{x}_{t_j}$ $(t_j = 1, \ldots, N_j)$ tracks of sensor i and j. The first step of the association is to compute the Λ likelihood of the common origin hypothesis, $\mathcal{H}_{t_i,t_j}$, relying on the following joint probability distribution function (pdf)

$$\Lambda\left(\mathcal{H}_{t_i,t_j}\right) = p\left(\hat{x}_{t_i}, \hat{x}_{t_j} | x_t\right) \tag{1}$$

conditioning the $\hat{x}_{t_i}$ and $\hat{x}_{t_j}$ local state estimates on the x_t actual state of the corresponding target. This can be rewritten by moving the estimate $\hat{x}_{t_i}$ into the conditioning as

$$\Lambda\left(\mathcal{H}_{t_i,t_j}\right) = p\left(\hat{x}_{t_j} | \hat{x}_{t_i}, x_t\right)\, p\left(\hat{x}_{t_i}, | x_t\right). \tag{2}$$

As the x_t actual state is unknown, a noninformative "diffuse" prior assumption is made, meaning that x_t does not contribute to the conditioning. Thus, function (1) yields

$$\Lambda\left(\mathcal{H}_{t_i,t_j}\right) = p\left(\hat{x}_{t_j} | \hat{x}_{t_i}\right)\, p\left(\hat{x}_{t_i}\right). \tag{3}$$

However, without conditioning $p(\hat{x}_{t_i})$ is also a diffuse pdf represented by a uniform distribution over the $\mathcal{V}$ state space with V volume. Therefore, the likelihood function is given as

$$\Lambda\left(\mathcal{H}_{t_i,t_j}\right) = \frac{1}{V}\, p\left(\hat{x}_{t_j} | \hat{x}_{t_i}\right). \tag{4}$$

In automotive applications, Gaussian assumptions are often made, under which

$$p(\hat{x}_{t_j} | \hat{x}_{t_i}) = \mathcal{N}\left(\hat{x}_{t_j}, \hat{x}_{t_i}, S_{t_i,t_j}\right) = \mathcal{N}\left(\hat{x}_{t_j} - \hat{x}_{t_i}, 0, S_{t_i,t_j}\right) \tag{5}$$

where the innovation-like covariance, S_{t_i,t_j}, is given as

$$S_{t_i,t_j} = P_{t_i} + P_{t_j} - P_{t_i,t_j} - \left(P_{t_i,t_j}\right)^{\top}. \tag{6}$$

The cross-covariance, P_{t_i,t_j}, of the tracks, can be given by a Lyapunov recursion relying on the Kalman filter parameters of the sensors [43]. Since, however, automotive sensor suppliers usually do not provide this information, as proposed in [23], it can be estimated relying on the Hadamard product of the track covariances

$$P_{t_i,t_j} \approx \rho \sqrt{P_{t_i} \odot P_{t_j}} \tag{7}$$

where ρ denotes the cross correlation factor that can be determined by Monte-Carlo simulation, but $\rho \approx 0.4$ was found a good approximation for most applications.

Then, the assignment problem is solved through an optimization. However, for numerical reasons, instead of maximizing the likelihood, the assignment cost is defined as

$$c_{t_i,t_j} = -\log\left(p\left(\hat{x}_{t_j} | \hat{x}_{t_i}\right)\right) = D_{t_i,t_j} + \log\left(|S_{t_i,t_j}|\right) \tag{8}$$

where D_{t_i,t_j} is the normalized squared distance computed as

$$D_{t_i,t_j} = \left(\hat{x}_{t_j} - \hat{x}_{t_i}\right)^{\top} \left(S_{t_i,t_j}\right)^{-1} \left(\hat{x}_{t_j} - \hat{x}_{t_i}\right). \tag{9}$$

Note that term V^{-1} is neglected in (8) as it is only a scaling constant. Now, solving the following optimization problem:

$$\min_{a_{t_i,t_j}} \sum_{t_i=0}^{N_i} \sum_{t_j=0}^{N_j} c_{t_i,t_j} a_{t_i,t_j}$$

$$\text{s.t.} \begin{cases} \sum_{t_i=0}^{N_i} a_{t_i,t_j} = 1 \\ \sum_{t_j=0}^{N_j} a_{t_i,t_j} = 1 \end{cases} \tag{10}$$

results in the $a_{t_i,t_j} \in \{0, 1\}$ assignment decision variables

$$a_{t_i,t_j} = \begin{cases} 1, & \text{if track } t_i \text{ is assigned to } t_j \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

The index $t_i = 0$ and $t_j = 0$ indicates the unassignment hypothesis with a user-defined $c_{t_i,0} = c_{0,t_j} = c_0/2$ cost. It should be noted that this integer linear programming problem, for the case of two sensors, can be solved by the Hungarian algorithm [27]. The generalization of the problem is given for multiple sensors in [25].

In a sensor-to-global architecture, the local sensor tracks are assigned to the global system tracks, repeated separately for each sensor, as shown in Fig. 1(b). This forms a similar two-class assignment between local and global tracks as in the case of two sensors in a sensor-to-sensor configuration. The main difference is in the likelihood function, which no longer requires diffuse prior assumption since having an $\hat{x}_t$ estimate with P_t covariance for the true state of a given target thanks to the memory of the $\mathbf{x}_t$ global tracks. Therefore, the likelihood of the hypothesis, $\mathcal{H}_{t \to t_i}$, indicating that the $\mathbf{x}_{t_i}$ local track of sensor i was generated by the t target is

$$\Lambda\left(\mathcal{H}_{t \to t_i}\right) = p\left(\hat{x}_{t_i} | \hat{x}_t\right) \tag{12}$$

which results in the same c_{t_i,t_j} assignment cost

$$c_{t,t_j} = \left(\hat{x}_t - \hat{x}_{t_i}\right)^{\top} \left(S_{t_i,t}\right)^{-1} \left(\hat{x}_t - \hat{x}_{t_i}\right) + \log\left(\left|S_{t_i,t}\right|\right) \tag{13}$$

and the covariance, $S_{t_i,t}$, is computed according to (6). Then, the state estimates and covariances of the tracks assigned to each other according to the association can be fused by one of the conventional algorithms [43].

B. Track Management

Track management is intended to maintain the global tracks, handling appearing and disappearing objects. Moreover, indirectly, it plays a crucial role in false alarm filtering. When a new track appears, it must be determined whether it is generated by a real target or it is a false track.

In a sensor-to-sensor architecture, a limited number of options are available for track management, as the fused local tracks directly map to global tracks due to the lack of memory. A naive approach assumes only those fused tracks correspond to real targets, which are confirmed by all sensors theoretically capable of detecting it according to their FoVs. However, this can result in misdetected objects. Another approach is to fuse the existence probability of local tracks, as in [52]. Then, the fused existence probability provides a good metric to decide whether the track is real or false.

In a sensor-to-global architecture, when a local track is not assigned to either of the existing global tracks, a new *tentative* track is initiated. Then, after a couple of cycles, the track gets *confirmed* if it is assumed to be generated by a real target. In multiobject tracking, this process is called track confirmation. Multiple track confirmation approaches can be found in the literature [46], which are also often used in track-to-track fusion. However, generally, probability-based track management arguably provides the best performance. Therefore, the proposed TMS also relies on this approach, estimating the existence probability of the fused tracks.

The existence probability of the local sensor tracks can also be fused in different ways. In multiobject tracking, IPDA [47] and JIPDA [48] algorithms are widely used to compute the existence probability of the tracks in a Bayesian manner. In [49], IPDA and JIPDA are also adapted for track-to-track fusion in a sensor-to-global configuration, considering the existence probability of local sensor tracks. However, it cannot work in a sensor-to-sensor architecture. DS belief fusion generalizes the Bayesian theory by enabling sensor-to-sensor existence probability fusion and offering more flexibility in modeling sensor uncertainties. Supposing the possible states, $X = \{\exists, \nexists\}$, of the tracks, i.e., they either exist (real target) or not exist (false), the power set, 2^X, formed by all possible subsets of X yields

$$2^X = \{\emptyset, \{\exists\}, \{\nexists\}, \{\exists, \nexists\}\} \tag{14}$$

where $\{\exists, \nexists\}$ represents the uncertain hypothesis, which is not interpreted in Bayesian theory. Then, the DST assigns an $m(\alpha)$ belief mass to each $\alpha \in 2^X$ hypotheses so that

$$m(\alpha) \in [0, 1] \tag{15}$$

$$m(\emptyset) = 0 \tag{16}$$

$$\sum_{\alpha \in 2^X} m(\alpha) = 1. \tag{17}$$

The basic belief assignment (BBA) of an $\mathbf{x}_{t_i}$ local track provided by sensor i, is given as

$$m_{t_i}(\{\exists\}) = p_i^{\text{track}}\left(\hat{x}_{t_i}\right)\, p_i^{\text{trust}}\, p\left(\exists \mathbf{x}_{t_i}\right) \tag{18}$$

$$m_{t_i}(\{\nexists\}) = p_i^{\text{track}}\left(\hat{x}_{t_i}\right)\, p_i^{\text{trust}}\left(1 - p\left(\exists \mathbf{x}_{t_i}\right)\right) \tag{19}$$

$$m_{t_i}(\{\exists, \nexists\}) = 1 - p_i^{\text{track}}\left(\hat{x}_{t_i}\right)\, p_i^{\text{trust}} \tag{20}$$

where $p(\exists \mathbf{x}_{t_i})$ denotes the existence probability of the local sensor track, $\mathbf{x}_{t_i}$. The tracking probability, p_i^{track}, describes how likely sensor i can detect and track an object with state $\hat{x}_{t_i}$. It can be expressed according to the FoV of the sensor and considering other phenomena as occlusion [55]. While p_i^{trust} denotes the user-defined trust of sensor i. If a sensor tends to provide false tracks, p_i^{trust} should have a lower value. The BBA of the misdetection case, i.e., when none of the tracks from sensor i is assigned to another $\mathbf{x}_{t_j}$ track, yields to

$$m_{t_i}(\{\exists\}) = 0 \tag{21}$$

$$m_{t_i}(\{\nexists\}) = p_i^{\text{track}}\left(\hat{x}_{t_i}\right)\, p_i^{\text{trust}} \tag{22}$$

$$m_{t_i}(\{\exists, \nexists\}) = 1 - p_i^{\text{track}}\left(\hat{x}_{t_i}\right) p_i^{\text{trust}}. \tag{23}$$

Now, the question is how to fuse the BBA of the different sources. Dempster's rule [51] provides an optimal operator to fuse belief constraints of different sources. Thus, having the BBA of two $\mathbf{x}_{t_i}$, $\mathbf{x}_{t_j}$ local tracks, the belief masses of their $\mathbf{x}_t$ fused global track are computed according to [51] as

$$m_t(\alpha) = \frac{\sum_{\beta \cap \gamma = \alpha} m_{t_i}(\beta) m_{t_j}(\gamma)}{1 - \sum_{\beta \cap \gamma = \emptyset} m_{t_i}(\beta) m_{t_j}(\gamma)}. \tag{24}$$

The fused belief mass of each $\alpha \in 2^X$ hypothesis can be computed using (24) according to the intersections of the set hypothesis pairs.

In the case of fusing the existence probabilities of a local and a global track in a sensor-to-global architecture, the a priori BBA $\hat{m}_t^k$ of the global track $\mathbf{x}_t$ at time k is given by the prediction from the cycle $k-1$ as

$$\hat{m}_t^k(\{\exists\}) = \gamma\, m_t^{k-1}(\{\exists\}) \tag{25}$$

$$\hat{m}_t^k(\{\nexists\}) = \gamma\, m_t^{k-1}(\{\nexists\}) \tag{26}$$

$$\hat{m}_t^k(\{\exists, \nexists\}) = 1 - \left(\hat{m}_t^k(\{\exists\}) + \hat{m}_t^k(\{\nexists\})\right) \tag{27}$$

where γ denotes the prediction weight, indicating the increased uncertainty ($m_t^k(\{\exists, \nexists\})$) about the existence of $\mathbf{x}_t$ due to the prediction. Therefore, γ is usually given proportionally to the time elapsed between the $k-1$th and kth cycles. Now, if an $\mathbf{x}_{t_i}$ local track of sensor i is assigned to the $\mathbf{x}_t$ global track, its BBA in (25)–(27) is updated by fusing it with the BBA of $\mathbf{x}_{t_i}$ in (18)–(20) according to (24). When none of the local tracks of sensor i is assigned to $\mathbf{x}_t$, it is fused with the BBA of misdetection in (21)–(23). However, it must be noted that the fused beliefs are not independent as the belief masses of $\mathbf{x}_t$ are recursively updated by the local sensor tracks. Using Dempster's rule in such cases has come under criticism [53], which is reflected in the proposed method in Section III-B.

Then, in both sensor-to-sensor and sensor-to-global configurations, the $p(\exists \mathbf{x}_t)$ Bayesian existence probability of the $\mathbf{x}_t$ fused global track can be computed as

$$p(\exists \mathbf{x}_t) = m_t(\{\exists\}) + \frac{1}{2} m_t(\{\exists, \nexists\}) \tag{28}$$

which serves as a good metric for track management to determine whether the object should be confirmed or deleted.

III. Proposed Methodology

In this article, a novel track-to-track fusion algorithm is proposed with a hybrid architecture, combining the sensor-to-sensor and sensor-to-global architectures, as illustrated in Fig. 2. It comprises three modules: the T2TA, state and covariance fusion, and global tracking and TMS, each detailed in the following sections. The proposed architecture is novel in that the T2TA assigns the *local* sensor tracks to each other as in a sensor-to-sensor architecture. Still, a separate tracking and TMS maintains a list *global* tracks with unique identifiers for common ADAS functions such as ACC and AEB. Thus, the global tracks are fed back to the T2TA with the aim of reducing false assignments. The local tracks assumed to be generated by the same target are then fused, creating the *meta* tracks, which are finally assigned to the global tracks, updating them. This allows us to perform a hybrid track management combining sensor-to-sensor and sensor-to-global existence probability estimation. Just as sensor-to-sensor configuration, this hybrid architecture also requires a proper synchronization of local sensor tracks, which is established by predicting their

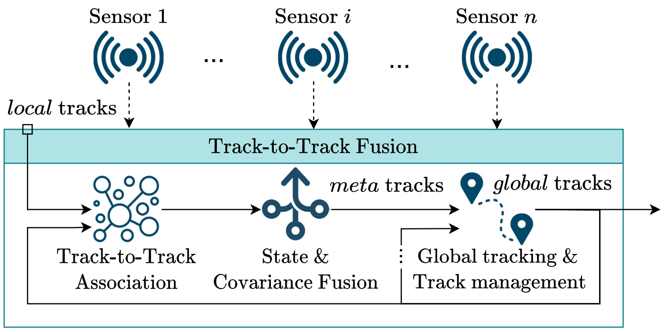

Fig. 2. Proposed hybrid track-to-track fusion architecture.

state and covariance to the joint fusion time. Further details of this synchronization method can be found in [31]. The main advantage of the proposed hybrid architecture is that the fusion can work synchronously, avoiding the aforementioned problems with asynchronous sensor updates. Still, it exploits the obvious advantages of sensor-to-global architecture.

A. Track-to-Track Association

In a sensor-to-sensor architecture, T2TA is intended to determine which local tracks of the sensors are generated by the same target and cluster them accordingly. Two goals are set with the proposed method: to obtain a computationally efficient, low-complex algorithm and still a robust association, compensating for false associations. The hybrid architecture shown in Fig. 2 allows us to replace the "diffuse" prior assumption of conventional association with an informative prior. In [28], the resulting assignments of the previous cycles are fed back to use as an informative prior, similar to the MHT tracker [29]. However, this solution can be computationally demanding. Therefore, instead, the feedback loop of the global tracks is exploited to replace the V^{-1} diffuse prior in the likelihood of the common origin hypothesis in (4)

$$\Lambda(\mathcal{H}_{t_i,t_j}) = p\left(\hat{x}^{\text{fus}}_{t_i,t_j}|\mathbf{X}\right)\, p\left(\hat{x}_{t_j}|\hat{x}_{t_i}\right) \tag{29}$$

where $\hat{x}^{\text{fus}}_{t_i,t_j}$ denotes the fused state estimate of $\hat{x}_{t_i}$ and $\hat{x}_{t_j}$ and $\mathbf{X} = \{\mathbf{x}_t\}_{t=1}^{N}$ is the set of global tracks. It means that $\hat{x}_{t_i}$ and $\hat{x}_{t_j}$ are fused tentatively according to Section III-B. It should be noted that for the sake of brevity, the k time index is neglected in (29) and the $\mathbf{x}_t$ global tracks of the previous, $k-1$th, cycle must be predicted to the current k time. Under Gaussian assumption, the informative prior in the proposed likelihood function (29) yields

$$p\left(\hat{x}^{\text{fus}}_{t_i,t_j}|\mathbf{X}\right) = \sum_{t=1}^{N} p\left(\hat{x}^{\text{fus}}_{t_i,t_j}|\hat{x}^{k|k-1}_{t}\right)\, p\left(\exists \mathbf{x}_t\right) + V^{-1} \tag{30}$$

$$p\left(\hat{x}^{\text{fus}}_{t_i,t_j}|\hat{x}^{k|k-1}_{t}\right) = \mathcal{N}\left(\hat{x}^{\text{fus}}_{t_i,t_j}, \hat{x}^{k|k-1}_{t},\; P^{\text{fus}}_{t_i,t_j} + P^{k|k-1}_{t}\right) \tag{31}$$

where $\hat{x}^{k|k-1}_{t}$ and $P^{k|k-1}_{t}$ denote the predicted state estimate and covariance of $\mathbf{x}_t$. The term, V^{-1}, in (30) indicates the birth density of $\hat{x}^{\text{fus}}_{t_i,t_j}$ if it does not correspond to either of the existing global tracks. It must be mentioned that the cross-covariance of $P^{\text{fus}}_{t_i,t_j}$ and $P^{k|k-1}_{t}$ is neglected here. Since (30) can be computationally demanding when the N total number of global tracks is high, the prior is estimated as

$$p\left(\hat{x}^{\text{fus}}_{t_i,t_j}|\mathbf{X}\right) \approx \sum_{\substack{\mathbf{x}_t \in \mathbf{X}^{\text{conf}}: \\ d(\hat{x}^{\text{fus}}_{t_i,t_j},\, \hat{x}^{k|k-1}_{t}) < d_{\text{gate}}}} p\left(\hat{x}^{\text{fus}}_{t_i,t_j}|\hat{x}^{k|k-1}_{t}\right) + V^{-1} \tag{32}$$

where $\mathbf{X}^{\text{conf}} \subseteq \mathbf{X}$ denotes the set of global tracks that have been already confirmed by the TMS, and d denotes the L^2 norm of $\hat{x}^{k|k-1}_{t}$ and $\hat{x}^{\text{fus}}_{t_i,t_j}$. The user-defined d_{gate} should be set considering that the higher values will result in higher computation cost; however, if d_{gate} is too low, the result will converge to the diffuse prior. Besides the "informative" prior, T2TA should consider a realistic sensor model. Therefore, the association likelihood in (29) is extended to consider the detection capability of the sensors as

$$\Lambda(\mathcal{H}_{t_i,t_j}) = p^{\text{track}}_{i}\left(\hat{x}^{i}_{t_i}\right)\, p^{\text{track}}_{j}\left(\hat{x}^{j}_{t_j}\right)\, p\left(\hat{x}^{\text{fus}}_{t_i,t_j}|\mathbf{X}\right)\, p\left(\hat{x}_{t_j}|\hat{x}_{t_i}\right) \tag{33}$$

where $p^{\text{track}}_{i}(\hat{x}_{t_i})$ denotes the probability that a track with $\hat{x}_{t_i}$ is being tracked by sensor i as an analogy of detection probability. Thus, if $p^{\text{track}}_{i}(\hat{x}_{t_i})$ is defined according to the FoV of the sensor, considering that they are unlikely to detect and track targets there, the assignment likelihood will be lower for tracks outside the sensor FoVs.

Having the revamped likelihood function, there is still an uncertain parameter of conventional methods, the gate likelihood, below which local tracks cannot be assigned to each other. When the assignment likelihoods are transformed into assignment costs, this parameter is called the cost of unassignment. This parameter highly impacts the performance of T2TA; therefore, our aim is to explicitly compute it with estimable sensor parameters. In [26], the gate likelihood is given by computing "no target" densities as the likelihood of the events that there is no target corresponding to the examined local tracks. The same logic is applied in the proposed T2TA, extending it to exploit the feedback of the global tracks. Therefore, the likelihood that there is no target for $\hat{x}_{t_i}$ local sensor track is defined as

$$\phi_{t_i} = \frac{\beta_i\, V_i^{-1}}{p\left(\hat{x}_{t_i}|\mathbf{X}\right)} + \frac{p\left(\hat{x}_{t_i}|\mathbf{X}\right)\, V_i^{-1}}{p\left(\hat{x}_{t_i}|\mathbf{X}\right) + p\left(\hat{x}^{\text{fus}}_{t_i,t_j}|\mathbf{X}\right)} + \kappa_i \tag{34}$$

where β_i and κ_i denote the birth and clutter (false track) densities of sensor i, and the likelihood, $p(\hat{x}_{t_i}|\mathbf{X})$, is computed as $p(\hat{x}^{\text{fus}}_{t_i,t_j}|\mathbf{X})$ in (32), replacing V with V_i that is the volume of the $\mathcal{V}_i$ state space of sensor i. The first term of (34) expresses how likely the $\mathbf{x}_{t_i}$ track is generated by a new target. The second term aims to consider spawning, quantifying how likely the $\mathbf{x}_{t_i}$ local track itself corresponds more to a target than the $\hat{x}^{\text{fus}}_{t_i,t_j}$ fused one. While the last term indicates whether $\mathbf{x}_{t_i}$ is a false track. Then, the gate likelihood, i.e., that $\mathbf{x}_{t_i}$ and $\mathbf{x}_{t_j}$ are not generated by the same target is given as

$$\Lambda(\mathcal{H}^{0}_{t_i,t_j}) = \phi_{t_i}\, \phi_{t_j}. \tag{35}$$

Hence, an individual gate likelihood is defined for each track pair. Moreover, likelihood values are usually small numbers. Therefore, for the sake of computational stability and to obtain a common assignment metric, the log-likelihood ratio of the assignment and gate likelihood is defined as

$$\begin{aligned}\lambda_{t_i,t_j} &= \log\left(\frac{\Lambda(\mathcal{H}_{t_i,t_j})}{\Lambda(\mathcal{H}^0_{t_i,t_j})}\right)\\ &= \frac{-D_{t_i,t_j} - \log(|S_{t_i,t_j}|)}{2}\\ &\quad - \log\left(\frac{\phi_{t_i}\,\phi_{t_j}\,2\pi^{n_x/2}}{p(\hat{x}^{\text{fus}}_{t_i,t_j}|\mathbf{X})\;p^{\text{track}}_i(\hat{x}_{t_i})\;p^{\text{track}}_j(\hat{x}_{t_j})}\right)\end{aligned} \tag{36}$$

where n_x denotes the dimension of the x state space. When $\Lambda(\mathcal{H}_{t_i,t_j})$ is greater than the gate likelihood, $\Lambda(\mathcal{H}^0_{t_i,t_j})$, the defined log-likelihood ratio is positive, otherwise negative. Thus, 0 is a universal assignment threshold of the proposed λ_{t_i,t_j} log-likelihood ratio, replacing the user-defined cost of unassignment. It should be noted that computing λ_{t_i,t_j} can be computationally heavy due to the evaluation of the Gaussian mixtures in (34). Hence, for the sake of runtime efficiency, it is only computed if the D_{t_i,t_j} normalized squared distance is smaller than D_{gate}, otherwise

$$\lambda_{t_i,t_j} \approx 0, \quad \text{if } D_{t_i,t_j} \geq D_{\text{gate}} \tag{37}$$

simplification is made. The parameter D_{gate} should be tuned with the same consideration as d_{gate} in (32), i.e., the higher the D_{gate}, the higher the computational demand of the T2TA; however, too low D_{gate} can spoil its robustness.

Now, the association can be formulated as an integer linear programming optimization problem, maximizing the log-likelihood ratios, similar to [25]. However, integer linear programming problems are generally NP-hard with high-computational costs. Moreover, these optimization models assume that the sensors provide at most one track of a target, preventing the assignment of tracks of the same sensor to each other. However, automotive sensors, particularly radars, sometimes provide duplicated tracks. Therefore, the assignment likelihood for tracks of the same sensor is defined as

$$\Lambda'\left(\mathcal{H}_{t_i,t'_i}\right) = p^{\text{dupl}}_i\,\Lambda\left(\mathcal{H}_{t_i,t_{i'}}\right), \quad \left(t'_i \neq t_i\right) \tag{38}$$

where p^{dupl}_i denotes the probability that the sensor i duplicates a target, which is an estimable sensor parameter. Thus, the log-likelihood ratio of the assignment between local tracks of the same sensor can be computed relying on (36)

$$\lambda'_{t_i,t'_i} = \log\left(\frac{\Lambda'\left(\mathcal{H}_{t_i,t'_i}\right)}{\Lambda(\mathcal{H}^0_{t_i,t'_i})}\right) = \lambda_{t_i,t'_i} + \log\left(p^{\text{dupl}}_i\right). \tag{39}$$

Then, the optimization problem is solved by a computationally efficient, iterative greedy clustering algorithm. Specifically, the clustering algorithm proposed in [31] is extended to tackle duplications, as shown in Algorithm 1. In each iteration, the $\{\mathbf{x}^\star_{t_{i\star}}, \mathbf{x}^\star_{t_{j\star}}\}$ pair of local tracks with the maximum log-likelihood ratio is searched. If none of these tracks belong to a cluster yet (line 23 of Algorithm 1), a new cluster will be formed with them as seen in line 25. The most significant difference from clustering in [31] is that the proposed one considers duplications, allowing the assignment of the tracks of the same sensor ($i^\star = j^\star$) to each other according to the log-likelihood ratio of duplication λ'_{t_i,t'_i} defined in (39). Thus, in the case when $i^\star = j^\star$, λ'_{t_i,t'_i} is also investigated as seen in line 24. If the track $\mathbf{x}^\star_{t_{i\star}}$ is already in a $\boldsymbol{c}^\star_{i\star}$ cluster (line 5), the log-likelihood ratios of the other $\mathbf{x}^\star_{t_{j\star}}$ track are analyzed with all $\mathbf{x}_{t_{ij}}$ tracks in $\boldsymbol{c}^\star_{i\star}$. Moreover, duplications are considered here as well. Specifically, if the reporting sensor ij of $\mathbf{x}_{t_{ij}} \in \boldsymbol{c}^\star_{i\star}$ is the same as that of $\mathbf{x}_{t^\star_{j\star}}$ ($ij = j\star$), the duplication likelihood $\lambda'_{t^\star_{j\star},t'_{j\star}}$ is taken into account. If each $\mathbf{x}_{t_{ij}}$ track of $\boldsymbol{c}^\star_{i\star}$ satisfies the log-likelihood ratio criteria, $\mathbf{x}^\star_{t_{j\star}}$ is appended to the cluster $\boldsymbol{c}^\star_{i\star}$. This is also a key difference compared to the algorithm in [31], as it does not investigate the likelihood with the other $\mathbf{x}_{t_{ij}}$ tracks of the cluster $\boldsymbol{c}^\star_{i\star}$ except $\mathbf{x}^\star_{t_{i\star}}$. Similarly, if $\mathbf{x}^\star_{t_{j\star}}$ is already in a cluster $\boldsymbol{c}^\star_{j\star}$ (line 10), $\mathbf{x}^\star_{t_{i\star}}$ is compared with all the tracks of $\boldsymbol{c}^\star_{j\star}$. If both $\mathbf{x}^\star_{t_{i\star}}$ and $\mathbf{x}^\star_{t_{j\star}}$ are in a cluster $\boldsymbol{c}^\star_{i\star}$ and $\boldsymbol{c}^\star_{j\star}$ (line 16), the log-likelihood ratio criterion is investigated for all sets of pair of tracks in $\boldsymbol{c}^\star_{i\star} \cup \boldsymbol{c}^\star_{j\star}$. If the log-likelihood

Algorithm 1 Track-to-Track Clustering

1: **given** λ_{t_i,t_j} assignment log-likelihood ratios, $i = 1, \ldots, n$, $j = 1, \ldots, n$, $t_i = 1, \ldots, N_i$, $t_j = 1, \ldots, N_j$
2: $\boldsymbol{C} = \emptyset$
3: **while** $\max(\lambda_{t_i,t_j}) > 0$ **do**
4: $t^\star_{i\star}, t^\star_{j\star} = \arg\max(\lambda_{t_i,t_j})$
5: **if** $\exists \boldsymbol{c}^\star_{i\star} \in \boldsymbol{C} : \mathbf{x}_{t^\star_{i\star}} \in \boldsymbol{c}^\star_{i\star}$ ($\mathbf{x}_{t^\star_{i\star}}$ in a cluster) **then**
6: **if** $\lambda_{t^\star_{j\star},t_{ij}} > 0 \quad \forall \mathbf{x}_{t_{ij}} \in \boldsymbol{c}^\star_{i\star} : ij \neq j^\star \wedge$
7: $\lambda'_{t^\star_{j\star},t'_{j\star}} > 0 \quad \forall \mathbf{x}_{t'_{j\star}} \in \boldsymbol{c}^\star_{i\star}$ **then**
8: $\boldsymbol{c}^\star_{i\star} = \boldsymbol{c}^\star_{i\star} \cup \mathbf{x}_{t^\star_{j\star}}$ (append)
9: **end if**
10: **else if** $\exists \boldsymbol{c}^\star_{j\star} \in \boldsymbol{C} : \mathbf{x}_{t^\star_{j\star}} \in \boldsymbol{c}^\star_{j\star}$ ($\mathbf{x}_{t^\star_{j\star}}$ in a cluster) **then**
11: **if** $\lambda_{t^\star_{i\star},t_{ij}} > 0 \quad \forall \mathbf{x}_{t_{ij}} \in \boldsymbol{c}^\star_{j\star} : ij \neq i^\star \wedge$
12: $\lambda'_{t^\star_{i\star},t'_{i\star}} > 0 \quad \forall \mathbf{x}_{t'_{i\star}} \in \boldsymbol{c}^\star_{j\star}$ **then**
13: $\boldsymbol{c}^\star_{j\star} = \boldsymbol{c}^\star_{j\star} \cup \mathbf{x}_{t^\star_{i\star}}$ (append)
14: **end if**
15: **else if** $\exists \boldsymbol{c}^\star_{i\star} \in \boldsymbol{C} : \mathbf{x}_{t^\star_{i\star}} \in \boldsymbol{c}^\star_{i\star} \wedge$
16: $\exists \boldsymbol{c}^\star_{j\star} \in \boldsymbol{C} : \mathbf{x}_{t^\star_{j\star}} \in \boldsymbol{c}^\star_{j\star}$ (both in a cluster) **then**
17: $\boldsymbol{c}^\star = \boldsymbol{c}^\star_{i\star} \cup \boldsymbol{c}^\star_{j\star}$
18: **if** $\lambda_{t_{ij},t_{ij'}} > 0 \quad \forall \{\mathbf{x}_{t_{ij}}, \mathbf{x}_{t_{ij'}}\} \in \boldsymbol{c}^\star : ij \neq ij' \wedge$
19: $\lambda'_{t_{ij},t'_{ij}} > 0 \quad \forall \{\mathbf{x}_{t_{ij}}, \mathbf{x}_{t'_{ij}}\} \in \boldsymbol{c}^\star$ **then**
20: $\boldsymbol{c}^\star_{i\star} = \boldsymbol{c}^\star_{i\star} \cup \boldsymbol{c}^\star_{j\star}$ (merge clusters)
21: $\boldsymbol{C} = \boldsymbol{C} \setminus \boldsymbol{c}^\star_{j\star}$
22: **end if**
23: **else** (none of them is in a cluster yet)
24: **if** $i^\star \neq j^\star \wedge \lambda'_{t^\star_{i\star},t^\star_{j\star}} > 0$ **then**
25: $\boldsymbol{C} = \boldsymbol{C} \cup \{\mathbf{x}_{t^\star_{i\star}}, \mathbf{x}_{t^\star_{j\star}}\}$ (new cluster)
26: **end if**
27: **end if**
28: $\lambda_{t^\star_{i\star},t^\star_{j\star}} = 0$
29: $\lambda_{t^\star_{i\star},t'_{j\star}} = 0 \quad \forall t'_{j\star} \in \{1, \ldots, N_{j\star}\}: \lambda_{t^\star_{i\star},t'_{j\star}} < \log(p^{\text{dupl}}_{j\star})$
30: $\lambda_{t'_{i\star},t^\star_{j\star}} = 0 \quad \forall t'_{i\star} \in \{1, \ldots, N_{i\star}\}: \lambda_{t'_{i\star},t^\star_{j\star}} < \log(p^{\text{dupl}}_{i\star})$
31: **end while**
32: **output** $\boldsymbol{C} = \{\boldsymbol{c}_{i_c}\}^{N_c}_{i_c=1} = \{\{\mathbf{x}^1_{t_i}\}^{n_1}_{t_i=1}, \ldots, \{\mathbf{x}^{N_c}_{t_i}\}^{n_{N_c}}_{t_i=1}\}$

ratio of each set of track pairs is greater than 0, the clusters are merged. If the investigated track pair of the two clusters is reported by the same sensor ($ij = ij'$), the duplication log-likelihood criterion is investigated. It is important to let clusters be merged, particularly over four sensors. Among duplication handling, neither cluster merging is considered in [31], which may result in false associations. Finally, besides $\lambda_{t_{i\star}^{\star},t_{j\star}^{\star}}$, the log-likelihood ratios of the tracks $\mathbf{x}_{t_{i\star}}^{\star}$ and $\mathbf{x}_{t_{j\star}}^{\star}$ with respect to the other tracks of the corresponding $i^{\star}$ and $j^{\star}$ sensors are all replaced with 0 in [31]. This step prevents $\mathbf{x}_{t_{i\star}}^{\star}$ from being assigned to other tracks of sensor $j\star$, assuming that sensors do not duplicate targets. In the proposed method, this is only done if the corresponding log-likelihood ratios are below $\log(p_{i\star}^{\mathrm{dupl}})$ and $\log(p_{j\star}^{\mathrm{dupl}})$, as seen in line 29 and 30 to let further assignments with them. This algorithm provides the $\{\boldsymbol{c}_{i_c}\}_{i_c=1}^{N_c}$ clusters comprising the $\{\mathbf{x}_{t_i}^{i_c}\}_{t_i=1}^{n_{i_c}}$ local sensor tracks that are assumed to be generated by the same target.

B. State and Covariance Fusion

After clustering the local tracks of the sensors, they must be fused to obtain a precise state estimate of their fused meta tracks. Specifically, the $\hat{x}_{t_i}^{i_c}$ state estimate and $P_{t_i}^{i_c}$ covariance of the $\{\mathbf{x}_{t_i}^{i_c}\}_{t_i=1}^{n_{i_c}}$ tracks in a $\boldsymbol{c}_{i_c}$ cluster are fused, determining the $\hat{x}_{i_c}^{\mathrm{fus}}$ state estimate and $P_{i_c}^{\mathrm{fus}}$ covariance of their fused meta track $\mathbf{x}_{i_c}^{\mathrm{fus}}$. This problem, i.e., state and covariance fusion, is well covered among the other modules of track-to-track fusion. Thus, a conventional method is implemented in the proposed track-to-track fusion. A detailed comparison of the conventional methods is even provided in [43], showing that the use of cross-covariance has the best performance compared to the covariance union and covariance intersection methods. Therefore, the cross-covariance method is used, which computes the fused state estimate $\hat{x}_{i_c}^{\mathrm{fus}}$ and covariance $P_{i_c}^{\mathrm{fus}}$ as

$$\hat{x}_{i_c} = \mathbf{W}_{i_c}^{\top} X_{i_c} = \mathbf{W}_{i_c}^{\top} \begin{bmatrix} \hat{x}_1^{i_c} & \dots & \hat{x}_{t_i}^{i_c} & \dots & \hat{x}_{n_{i_c}}^{i_c} \end{bmatrix}^{\top} \tag{40}$$

$$P_{i_c}^{\mathrm{fus}} = \mathbf{W}_{i_c}^{\top} P_{i_c} \mathbf{W}_{i_c} \tag{41}$$

where

$$\mathbf{W}_{i_c} = P_{i_c}^{-1} A_{i_c}^{\top} \left(A_{i_c} P_{i_c}^{-1} A_{i_c}^{\top} \right)^{-1} \qquad \mathbf{W}_{i_c} \in \mathbb{R}^{n_{i_c} n_x \times n_x} \tag{42}$$

$$A_{i_c} = \mathbf{1}_{n_{i_c}} \otimes I_{n_x} \qquad \mathbf{1}_{n_{i_c}} \in \{1\}^{n_{i_c}}, I_{n_x} \in \{0,1\}^{n_x \times n_x} \tag{43}$$

$$P_{i_c} = \begin{bmatrix} P_1^{i_c} & \dots & P_{1,t_i}^{i_c} & \dots & P_{1,n_{i_c}}^{i_c} \\ & & \vdots & & \\ P_{t_i,1}^{i_c} & \dots & P_{t_i}^{i_c} & \dots & P_{t_i,n_{i_c}}^{i_c} \\ & & \vdots & & \\ P_{n_{i_c},1}^{i_c} & \dots & P_{n_{i_c},t_i}^{i_c} & \dots & P_{n_{i_c}}^{i_c} \end{bmatrix}. \tag{44}$$

$\mathbf{1}_{n_{i_c}}$ and I_{n_x} denoting the vector comprising n_{i_c} ones, and the $n_x \times n_x$ identity matrix, respectively. The $P_{t_i,t_j}^{i_c}$ cross-covariance matrices in the diagonal elements of P_{i_c} in (44) are computed according to the assumption in (7).

This method is also used in the T2TA to tentatively fuse $\mathbf{x}_{t_i}$ and $\mathbf{x}_{t_j}$ local tracks, determining the fused state $\hat{x}_{t_i,t_j}^{\mathrm{fus}}$, and covariance $P_{t_i,t_j}^{\mathrm{fus}}$ in (36). However, for two tracks, it can be simplified to

$$\hat{x}_{t_i,t_j}^{\mathrm{fus}} = \hat{x}_{t_i} + \chi \left(\hat{x}_{t_j} - \hat{x}_{t_i} \right) \tag{45}$$

$$P_{t_i,t_j}^{\mathrm{fus}} = P_{t_i} - \left(P_{t_i} - P_{t_i,t_j} \right) S_{t_i,t_j}^{-1} \left(P_{t_i} - P_{t_i,t_j} \right)^{\top} \tag{46}$$

where

$$\chi = (P_{t_i} - P_{t_i,t_j}) S_{t_i,t_j}^{-1} \tag{47}$$

calculating the innovation covariance S_{t_i,t_j} and the cross-covariance P_{t_i,t_j} according to (6) and (7), respectively.

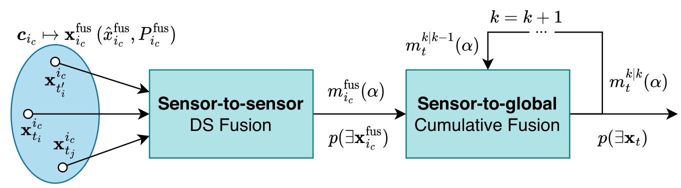

Fig. 3. Proposed two-step track existence probability fusion.

C. Global Tracking and Track Management

Fusing the state and covariance of the local sensor tracks in the same cluster determines the so-called meta tracks, as seen in Fig. 2. As mentioned above, many ADAS functions, such as AEB and ACC, require the input tracks to be referable with unique identifiers. Since, however, meta-tracks are created from the resulting clusters of T2TA with varying elements, they do not have unique identifiers. Therefore, an additional tracking algorithm is implemented in the last step of the proposed architecture to maintain a list of global tracks with unique identifiers. This also contributes to a more sophisticated track management and facilitates the hybrid track-to-track architecture. Since one of the main targets of this research was to obtain a low-complex fusion algorithm, a runtime-efficient tracking algorithm in one of our previous works [56] is extended with the proposed TMS. As existence probability track management usually outperforms other approaches (e.g., M/N logic), the proposed method also relies on the existence probabilities of the fused tracks. This requires the fusion of the existence probabilities of local sensor tracks. It has to be mentioned that algorithms tackling this problem already exist, as discussed in Section II-B. In multiobject tracking, existence probability estimation is usually integrated with data association, such as in IPDA [47] and JIPDA [48], which update the existence probability of the tracks objects in a Bayesian manner. However, in track-to-track fusion, local sensor tracks usually already have existence probability estimates. IPDA and JIPDA are adapted in [49] to consider these estimates of fused tracks. Aeberhard et al. proposed a method in [52] relying on DST as a generalization of the Bayesian estimation to fuse the existence probabilities of local sensor tracks. This method is also adapted to a sensor-to-global architecture in [21] due to the substantial advantages over the sensor-to-sensor strategy.

However, DS fusion assumes that the fused sources are independent belief constraints [53], which clearly does not hold when fusing the beliefs of a global track with its corresponding local track. Therefore, a revamped, two-step existence probability fusion method is proposed, illustrated in Fig. 3 that combines the sensor-to-sensor and sensor-to-global

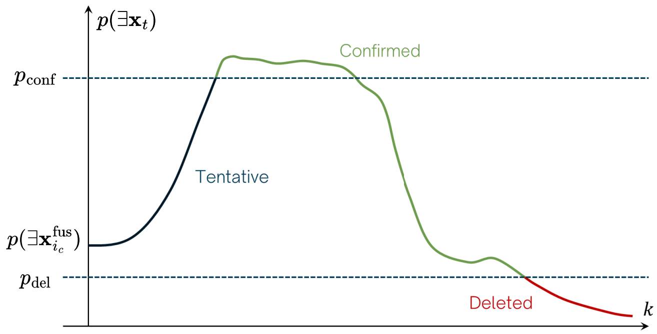

Fig. 4. Illustration of existence probability-based track management.

strategy, exploiting the hybrid architecture. First, the existence probabilities of the local tracks in the $\boldsymbol{c}_{i_c}$ cluster are fused in a sensor-to-sensor manner, according to the DS rule in (24), as in [52]. The output of the first step comprise the $m_{i_c}^{\text{fus}}(\alpha)$ fused beliefs for each $\alpha \in 2^X$ hypothesis, and the $p(\exists \mathbf{x}_{i_c}^{\text{fus}})$ fused existence probability of the meta track $\mathbf{x}_{i_c}^{\text{fus}}$. In the second step, the beliefs $m_{i_c}^{\text{fus}}(\alpha)$ of the meta tracks are fused with the $m_t^{k|k-1}(\alpha)$ a priori beliefs of the $\mathbf{x}_t$ global track. This step performs a sensor-to-global fusion, which contributes to the continuity of the tracks. The assignments between the meta and global tracks are made according to the runtime efficient nearest neighbor (NN) algorithm proposed in [56]. The a priori mass beliefs $m_t^{k|k-1}(\alpha)$ are computed by predicting the $m_t^{k-1|k-1}(\alpha)$ beliefs from the previous $k-1$th time as in (25)–(27). Reflecting the criticism of using Dempster's rule in situations when the sources are not independent belief constraints, the cumulative fusion rule [57] is implemented. Thus, the a priori beliefs $m_t^{k|k-1}(\alpha)$ of $\mathbf{x}_t$ are updated accordingly as

$$m_t^k(\{\exists\}) = \frac{m_{i_c}^{\text{fus}}(\{\exists\})\,\hat{m}_t^k(\{\exists,\nexists\}) + m_{i_c}^{\text{fus}}(\{\exists,\nexists\})\,\hat{m}_t^k(\{\exists\})}{C} \tag{48}$$

$$m_t^k(\{\nexists\}) = \frac{m_{i_c}^{\text{fus}}(\{\nexists\})\,\hat{m}_t^k(\{\exists,\nexists\} + m_{i_c}^{\text{fus}}(\{\exists,\nexists\})\,\hat{m}_t^k(\{\nexists\}))}{C} \tag{49}$$

$$m_t^k(\{\exists,\nexists\}) = \frac{m_{i_c}^{\text{fus}}(\{\exists,\nexists\})\,\hat{m}_t^k(\{\exists,\nexists\})}{C} \tag{50}$$

where the normalizing factor C is

$$C = m_{i_c}^{\text{fus}}(\{\exists,\nexists\}) + \hat{m}_t^k(\{\exists,\nexists\}) - m_{i_c}^{\text{fus}}(\{\exists,\nexists\})\,\hat{m}_t^k(\{\exists,\nexists\}). \tag{51}$$

After updating the belief masses of the global track $\mathbf{x}_t$ according to (48)–(50), its Bayesian existence probability $p(\exists \mathbf{x}_t)$ is computed as in (28). Then, the track management is performed relying on this Bayesian existence probability, which is illustrated in Fig. 4. When a new global track appears, its $p(\exists \mathbf{x}_t)$ probability is initialized by the existence probability $p(\exists \mathbf{x}_{i_c}^{\text{fus}})$ of the corresponding $\mathbf{x}_{i_c}^{\text{fus}}$ meta track. Then, $p(\exists \mathbf{x}_t)$ is updated recursively according to the two-step existence probability estimation. If the tentative global track is updated in the consecutive cycles by a $\mathbf{x}_{i_c}^{\text{fus}}$ meta track with high $p(\exists \mathbf{x}_{i_c}^{\text{fus}})$ existence probability, $p(\exists \mathbf{x}_t)$ starts increasing as shown in Fig. 4. When $p(\exists \mathbf{x}_t)$ exceeds the predefined p_{conf} confirmation threshold, the global track $\mathbf{x}_t$ gets confirmed and transmitted to the ADAS application layer. Then, when the track is misdetected by multiple sensors, its $p(\exists \mathbf{x}_t)$ existence probability starts decreasing, as seen in Fig. 4. Still, the track is kept confirmed until its $p(\exists \mathbf{x}_t)$ existence probability drops below the p_{del} deletion threshold when it is removed from the track list. The state estimate $\hat{x}_t$ and P_t covariance of the global track $\mathbf{x}_t$ is directly updated with $\hat{x}_{i_c}^{\text{fus}}$ and $P_{i_c}^{\text{fus}}$ without any additional filtering.

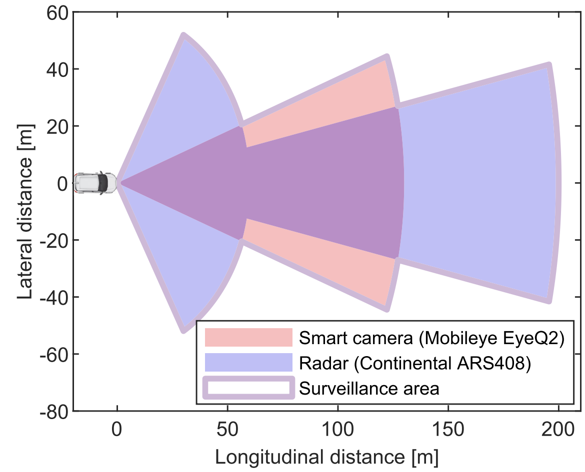

Fig. 5. Illustration of the evaluation sensor cluster.

IV. Experimental Setup

The evaluation is performed on a real-world measurement of a conventional ADAS perception system depicted in Fig. 5 comprising an automotive radar and smart camera. However, it should be mentioned that the proposed methodology is generic and extendable with arbitrary sensors providing tracked objects, as seen in Fig. 2. The radar provides the tracked objects with 12 Hz frequency up to 200 m under an 18° far-range opening angle. It also has a near scan zone with 120° FoV angle up to 50 m. The camera has a 40° FoV with ~120 m view range, providing its tracked objects with 9 Hz frequency. It must be noted that although it tracks more objects internally, at most, four tracks are provided by the camera due to communication limitations. While the maximum number of radar tracks is 50. The measurement was taken on the Hungarian M1/M7 highway under nice weather and average workday afternoon traffic conditions. Since the raw images of the smart camera are inaccessible, the measurement setup is extended for the evaluation with a monocular GoPro camera synchronized with the radar and smart camera data. A 7000-frame (~116 s, 60 frames/s) section of the measurement was selected for the evaluation so that it includes the heaviest traffic and dynamic scenarios, such as frequent lane changes and standstill objects in the emergency lane. The ground truth of this scene is generated semi-automatically using [58]. However, it should be noted that the evaluation is performed up to 90 m, as beyond this, the depth uncertainty of the monocular camera is no longer sufficient to generate ground truth. The

TABLE I
SENSOR PARAMETERS USED IN THE FUSION ALGORITHMS

Parameter	Notation	Smart camera	Radar
Tracking probability	p^{track}	0.85	0.95
Trust probability	p^{trust}	0.99	0.85
Duplication probability	p^{dupl}	10^{-9}	10^{-9}
State space volume	V	1.47×10^{7}	2.62×10^{7}
Birth density	β	6.78×10^{-8}	3.82×10^{-8}
Clutter density	κ	6.78×10^{-11}	3.82×10^{-7}

ground truth comprises 34 174 annotations of 28 individual objects, including passenger cars, trucks, and buses, as shown in the following video: https://shorturl.at/ixhma.

The sensors provide the input tracks estimating the state of the surrounding objects as

$$\hat{x} = \begin{bmatrix} d_x & v_x & d_y & v_y \end{bmatrix}^\top \tag{52}$$

including the relative position, i.e., distance, and velocity with d_x, v_x longitudinal and d_y, v_y lateral components, respectively. The sensors only provide the variances of the state variables of $\hat{x}$; hence, the P covariance of the input tracks is modeled with diagonal matrices. The parameters of the sensors used in the proposed method are collected in Table I. Although the tracking probability p^{track} is usually given according to the sensor FoV, for the sake of simplicity, every parameter is set constant as it does not distort the comparison of the proposed fusion with conventional methods. It should be noted that the β birth and κ clutter densities are computed relying on the V state space volume as

$$\beta = V^{-1}\,\lambda_b\,, \quad \kappa = V^{-1}\,\lambda_c \tag{53}$$

assuming Poisson point processes [59], where λ_b and λ_c denote the Poisson rate (mean value) of the number of birth and false tracks. The gate parameters, $d_{\text{gate}} = 10$ and $D_{\text{gate}} = 25$, of the proposed methodology are set to make a balance between performance and runtime. The proposed T2TA is compared with conventional methods, requiring the cost of unassignment parameter c_0, defined according to the formula in [26]. The confirmation threshold p_{conf} and deletion threshold p_{del} parameters of the TMS are set 0.9 and 0.2, considering false track filtering and continuous object tracking.

The proposed track-to-track fusion algorithm is evaluated according to two aspects: its performance and computational efficiency. The performance is evaluated according to common object detection and multiobject tracking metrics such as precision, recall, F1, and MOTA scores. First, the confirmed global tracks provided by the fusion are assigned to the ground-truth objects using the Kuhn–Munkres algorithm [27] within $d_c^{\text{lon}} = 10$ m longitudinal and $d_c^{\text{lat}} = 1.5$ m lateral cutoff distance. Each assignment between the output tracks and the ground-truth objects is counted as a true positive, while the tracks not assigned to either of the ground-truth objects are assumed to be false positives. The ground-truth objects without tracks in their cutoff distance are noted as false negatives. Then, the precision and recall of the fusion are computed as

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{54}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{55}$$

expressing the false detection filtering and detection performances, respectively, where TP, FP, and FN denote the total number of true positives, false positives, and false negatives. The F1 score is the harmonic mean of precision and recall, indicating the overall object detection performance of the fusion. These are common object detection metrics [60]. While the MOTA score provides a conventional object-tracking performance [61], considering the track ID consistency too as

$$\text{MOTA} = 1 - \frac{\text{FN} + \text{FP} + \text{IDS}}{\text{TP} + \text{FN}} \tag{56}$$

where IDS denotes the total number of track ID changes. The computational efficiency is evaluated by the runtime complexity of the T2TA over the number of tracks, as the computational demand of other modules of the fusion is negligible compared to the T2TA. The evaluation is performed on a Lenovo ThinkCentre PC with an Intel Core i7-10700 2.9 GHz processor and 16 GB memory in MATLAB R2021b.

V. RESULTS

The proposed methodology is compared with conventional methods, focusing separately on the T2TA, TMS, and overall fusion performance. Therefore, the evaluation results of these are detailed in the following sections. Video illustrations showcasing the results of the compared algorithms are available at https://shorturl.at/KPOGf.

A. Track-to-Track Association

First, the proposed T2TA is compared with the following existing methods.

1) The optimization-based [25], which formulates the association as an integer linear programming problem relying on the conventional T2TA negative log-likelihood assuming "diffuse" prior.
2) A low-complex iterative method [31], which solves the assignment problem based on a greedy clustering algorithm also using the conventional assignment costs.
3) The sensor-to-global approach [30], which assigns the local sensor track to the global tracks sequentially for each sensor using the Hungarian method.

These methods are evaluated by replacing the proposed T2TA algorithm with them but keeping the rest of the fusion, including the TMS. The resulting performance metrics are shown in Table II. As seen, the proposed T2TA outperforms the other conventional methods due to its highest precision. Although the recall of the optimization-based and the iterative methods are higher than of the proposed one, the 0.07% difference is negligible compared to the precision gain over 1%, which then results in the highest F1 and MOTA scores as overall object detection and tracking metrics. The high precision of the proposed method is explained by the duplication handling, while the recall is maintained by the revamped likelihood

TABLE II
PERFORMANCE COMPARISON OF T2TA METHODS

Method	Precision	Recall	F1 Score	MOTA
Optimization-based [25]	95.32%	**86.14%**	90.50%	81.90%
Iterative [31]	95.15%	**86.14%**	90.42%	81.74%
Sensor-to-global [30]	95.76%	84.42%	89.73%	80.66%
Proposed	**96.55%**	86.07%	**91.01%**	**82.98%**

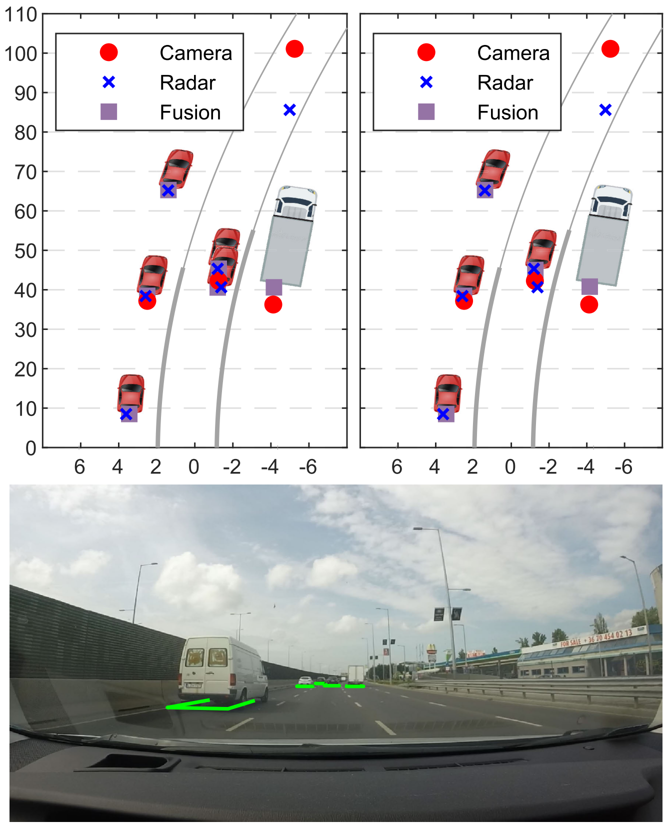

Fig. 6. Qualitative comparison of the proposed T2TA (right) and the existing iterative (left) [31], illustrating duplication handling.

function using informative prior. The effect of duplication handling is also illustrated in Fig. 6, showing that the iterative clustering in [31] cannot fuse the duplicated tracks of the radar, creating a false track in the ego lane. Although the sensor-to-global assignment has slightly better precision than the two other conventional methods, its recall falls short compared to them, having the lowest F1 and MOTA scores as well.

The other crucial aspect of the proposed T2TA is its computational complexity, which is given separately for the two steps: the computation of log-likelihood ratios and the clustering itself. The complexity of the proposed log-likelihood ratio computation is illustrated by its average runtime over the number of input tracks in Fig. 7, comparing it with the complexity of the conventional assignment cost computation in [25]. Although the runtime of the conventional method seems to be linear, it takes time $\mathcal{O}(N_r \times N_c)$, where N_r and N_c denotes the number of local tracks of the radar and camera. Since the camera can provide at most four tracks, the runtime seems to have a $\mathcal{O}(N_r)$ complexity. The proposed

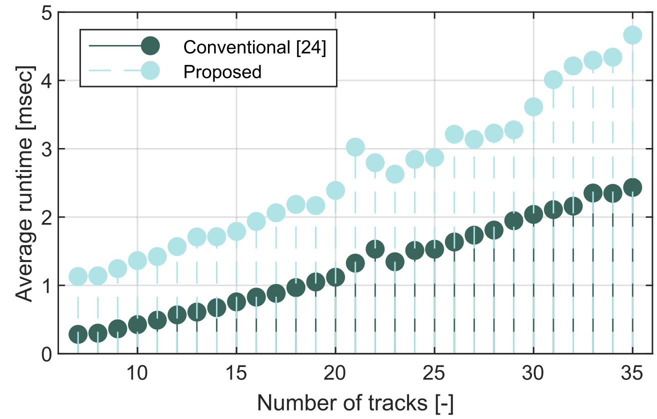

Fig. 7. Runtime complexity comparison of the different T2TA algorithms concerning the assignment cost/log-likelihood ratio computation.

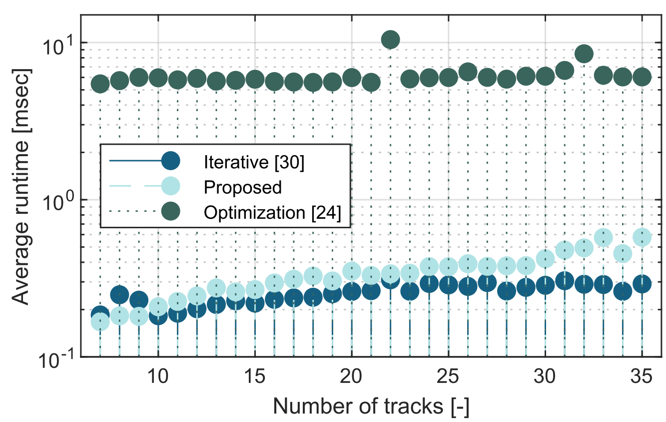

Fig. 8. Runtime complexity comparison of the different T2TA algorithms concerning the clustering (solving the assignment problem).

log-likelihood ratio computation has a slightly higher runtime due to the evaluation of the Gaussians in (30) and (34) and since it is not only performed for track pairs of different sensors. Still, its $\mathcal{O}(N^2)$ complexity, where N denotes the total number of sensor tracks, is favorable and allows real-time application despite the overhead compared to [25]. The complexity of the other step of T2TA, i.e., solving the assignment problem, defined by the assignment costs or the log-likelihood ratios, is shown in Fig. 8 on a logarithmic scale for visibility. The proposed clustering in the algorithm is compared with the iterative clustering in [31] and with the integer linear programming optimization method in [25]. It should be noted that the sensor-to-global association is neglected here, as its runtime also scales with the number of global tracks. As seen, the optimization-based clustering has the highest runtime which seems to be independent of the number of tracks. Still, in dense traffic, when the targets are close to each other, the runtime peak can be significantly high, as seen in the peak at $N = 22$, risking the real-time applicability. The proposed clustering has a similar favorable complexity as the iterative clustering in [31]. Therefore, it matches the research goal of targeting a low-complex but robust T2TA, which allows real-time applications. Although these runtimes were measured in an offline MATLAB environment, it should be mentioned that we deployed the proposed fusion on a Speedgoat Baseline real-time target on which it was running with a 5% peak load.

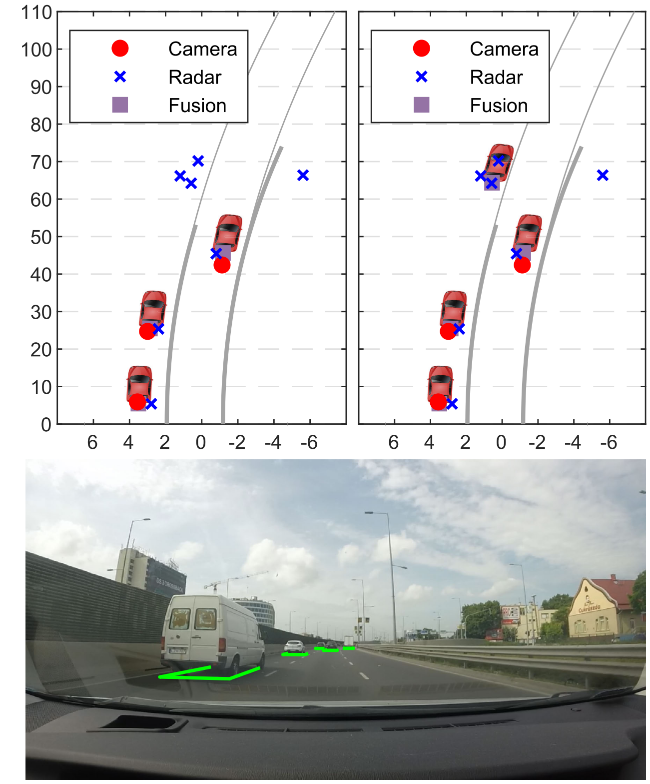

Fig. 9. Qualitative comparison of the proposed TMS (right) and the existing DS S2G (left) [21], illustrating misdetection.

B. Track Management System

The performance of the proposed TMS is compared with the methods rely on the following.

1) Assignment history of the global tracks (M/N logic) [30].
2) Existence probability of the tracks, fusing the existence probability of the local tracks using the DS method in a sensor-to-sensor configuration (DS S2S) [52].
3) Existence probability of the tracks, fusing the existence probabilities using the DS method in a sensor-to-global configuration (DS S2G) [21].

The comparison is performed by replacing the TMS of the proposed track-to-track fusion (keeping the T2TA) and investigating the effect of the different methods. Then, the fusion result is evaluated according to Section IV, computing the performance metrics given in Table III. Although the M/N logic has the highest recall, its 52.81% precision results in the lowest overall scores, i.e., 67.78% F1 and 9.99% MOTA score. The low precision is explained by the false tracks provided by the radar, which cannot be efficiently filtered by the M/N logic. From an ADAS application point of view, this can result in dangerous false warnings or actuation. In contrast, the DS existence probability fusion in a sensor-to-sensor configuration has the highest, 99.31%, precision, efficiently filtering the false positives. However, consequently, it misses many objects, having a lower 67.34% recall. Still, it's over 80% F1 score is significantly better than that of the M/N logic. Since in sensor-to-sensor architecture, the global tracks are not maintained with unique identifiers, the MOTA score considering track ID changes is not interpreted for this method. Among the existing

TABLE III
PERFORMANCE COMPARISON OF TMS METHODS

Method	Precision	Recall	F1 Score	MOTA
M/N logic [30]	52.81%	**94.58%**	67.78%	9.99%
Dempster-Shafer S2S [52]	**99.31%**	67.34%	80.26%	N/A
Dempster-Shafer S2G [21]	97.81%	77.33%	86.38%	75.59%
Proposed	96.55%	86.07%	**91.01%**	**82.98%**

TABLE IV
PERFORMANCE COMPARISON OF THE PROPOSED FUSION WITH THE SENSORS, OTHER FUSION (CENTRAL TRACKING) ALGORITHMS

	Precision	Recall	F1 Score	MOTA
Radar	48.75%	**87.06%**	62.50%	-4.75%
Camera	**97.43%**	73.40%	83.73%	71.46%
JIPDA [48]	95.37%	86.73%	90.85%	82.52%
Robust GM-PHD [62]	96.81%	84.85%	90.44%	82.05%
Proposed	96.55%	86.07%	**91.01%**	**82.98%**

TMS methods, applying DS in a sensor-to-global architecture performs arguably the best with an 86.38% F1 score and 75.59% MOTA score. However, its recall falls significantly short compared to the proposed two-step existence probability fusion. The explanation for this is that DS underestimates the existence probability of the global tracks when they are misdetected by one of the sensors. This can result in completely missed objects, as the farther object in the left lane illustrated in Fig. 9. Meanwhile, the proposed two-step fusion can efficiently resolve these errors.

C. Overall Fusion

Finally, the performance of the overall proposed track-to-track fusion is compared with the sensors and other state-of-the-art object fusion methods relying on central multiobject tracking algorithms, comprising:

1) a robust GM-PHD tracker in of our previous works [62]; and
2) one of the widespread trackers, the JIPDA [48],

computing the same metrics. The results are given in Table IV, highlighting the challenges of the sensor setup. Although the radar can detect most of the relevant objects, it provides numerous false tracks generated by static objects, such as guardrails, traffic signs, and highway overpasses, resulting in low precision and a negative MOTA score. Meanwhile, the camera has high precision but a lower recall. As seen in Table IV, the proposed track-to-track fusion algorithm slightly outperforms other fusion methods relying on a central tracking algorithm. Although the JIPDA tracker has a higher 86.73% recall than the proposed one, its precision falls short by more than 1%, resulting in lower overall metrics. Despite the <0.5% F1 and MOTA score difference, it must be mentioned that in ADAS applications, precision is more important than recall due to false alarm filtering. Moreover, JIPDA has a high-computational cost, having a combinatorial complexity [62]. The precision of the robust GM-PHD tracker in [62] is greater by 0.26%. Still, its F1 and MOTA scores are lower by more than 0.5% due to the 1.22% recall difference. Although the camera has the highest precision, indicating that it provides the least false tracks, which is an important aspect from an ADAS perspective. However, it provides at most four tracks due to control area network (CAN) interface limits, resulting in a lower recall, indicating that it has a worse detection performance. Meanwhile, the proposed track-to-track fusion makes a good balance between false filtering (precision) and object tracking performance (recall), resulting in the highest overall metrics. Therefore, it is robust against such sensor uncertainties as false tracks and misdetections.

VI. Conclusion and Future Work

This article proposed a robust and low-complex fusion algorithm based on a novel hybrid track-to-track architecture. The proposed architecture allows us to introduce the revamped assignment likelihood function using informative prior, replacing the "diffuse" prior assumption of conventional methods. Moreover, the proposed likelihood function considers duplicated targets, also providing an exact computation of gate likelihood relying on common sensor parameters. Finally, an iterative algorithm clusters the local sensor tracks with low-computational cost, also considering duplicates. On top of that, a TMS is also provided, relying on a revamped two-step existence probability fusion, reflecting the criticism of the DS belief fusion. The proposed T2TA and TMS outperform the conventional methods, as does the overall fusion, surpassing other state-of-the-art fusion methods that rely on central tracking algorithms.

In future work, the proposed fusion approach will be enhanced by incorporating a more advanced object birth model that accounts for the sensor FoV, thereby expediting track confirmation. Additionally, we plan to develop an adaptive clutter model to further improve the efficiency of false alarm filtering. The current evaluation accounts for the robustness of the proposed fusion against false tracks and misdetections. Hence, a more comprehensive robustness analysis will be conducted in future work for scenarios such as sensor faults (e.g., communication loss) or external attacks.

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